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Electro-Magnetic Field (force caused by potential)

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m = \frac{1}{c} \left(\frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left(cp_{0_{dipol}} + m_{0_{dipol}} \times \frac{r}{|r|} \right) \stackrel{\text{Near Field}}{\approx} \frac{1}{2} \frac{1}{2\pi} \mu_0 q \frac{v \times r}{|r|^2} \quad (1)$$

Magnetic Field in Matter

Excitation, Displacement

$$\mathcal{H} = \dots \quad (2)$$

Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservative}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} - \frac{v}{c}\right)\right]}{r^2}}_{\text{acceleration}} \right] \quad (3)$$

Electric Field in Matter

Excitation, Displacement

$$\mathcal{D} = \dots \quad (4)$$